

INSIGHTS DOCUMENT

US Disaster Declarations Analysis

◆ Milestone-1: Data Cleaning & Exploration Insights

The initial dataset contained multiple columns with inconsistent formats and unnecessary information. During preprocessing, irrelevant columns were removed and important columns such as **state, incident type, and declaration date** were retained.

The date column was converted into a proper datetime format, which enabled extraction of **year and month features** for further analysis.

Handling missing values ensured that the dataset was clean and reliable for analysis. This step was critical because inaccurate or incomplete data could lead to incorrect conclusions.

Overall, this milestone ensured that the dataset was **structured, consistent, and ready for analysis**, forming the foundation for all subsequent steps.

◆ Milestone-2: Temporal Analysis Insights

The temporal analysis revealed that disaster declarations have **increased significantly over time**, especially after the year 2000.

This trend suggests that either:

- natural disasters are becoming more frequent
- reporting and declaration systems have improved
- environmental and climate changes are influencing disaster patterns

The **yearly trend graph** showed fluctuations, but the overall pattern indicates an upward trend.

The **monthly analysis** revealed that disasters are not evenly distributed across the year. Certain months show higher disaster occurrences, indicating **seasonal patterns** such as hurricane seasons and heavy rainfall periods.

The **rolling average** helped smooth out short-term fluctuations and confirmed the long-term increasing trend.

Overall, this milestone highlighted that **time plays a significant role in disaster occurrence patterns**.

◆ Milestone-3: Geographical Analysis Insights

The geographical analysis showed that disaster occurrences are **not evenly distributed across states**.

Certain states such as:

- Texas
- California
- Florida

consistently show higher disaster declarations.

This can be attributed to:

- large geographic size
- exposure to multiple disaster types
- coastal and climatic conditions

The **choropleth map** clearly highlighted disaster hotspots, with darker regions indicating higher disaster frequency.

The **state comparison chart** confirmed that southern and coastal states experience more disasters.

The **incident type distribution across states** showed that different regions are affected by different disaster types, indicating strong geographic influence.

Overall, this milestone demonstrated that **location is a key factor in disaster patterns**.

◆ Milestone-4: Incident Type Analysis Insights

The incident type analysis revealed that certain disaster types occur more frequently than others.

The most common disaster types include:

- Severe Storms
- Floods
- Hurricanes

This indicates that **weather-related disasters dominate in the United States**.

The **state vs incident heatmap** showed that specific disaster types are concentrated in certain states.

For example:

- Coastal states show higher hurricane occurrences
- Some regions show higher flood or storm activity

The **stacked bar chart** highlighted how different disaster types contribute to total disasters in each state, showing clear regional variation.

The **assistance program analysis** revealed that disasters such as hurricanes and floods trigger higher assistance support.

It was also observed that:

- Public assistance programs are used more frequently
- Large-scale disasters require more government support

Overall, this milestone showed that **disaster type plays a major role in understanding disaster impact and response**.

◆ Combined Insights (Final Understanding)

By combining insights from all milestones, several key patterns emerge:

1. Disaster frequency has increased over time, indicating changing environmental conditions.
2. Disaster occurrences are highly dependent on geographic location, with certain states experiencing more frequent disasters.
3. Weather-related disasters such as storms, floods, and hurricanes dominate across the United States.
4. Different regions experience different types of disasters based on their geographic characteristics.
5. Assistance programs are more frequently activated for large-scale disasters, especially in high-risk regions.

Overall, disaster patterns are influenced by a combination of:

- Time (increasing trend)
- Geography (state-wise variation)
- Disaster type (weather dominance)

These insights provide a comprehensive understanding of disaster behavior and can help in better disaster preparedness and planning.